

Research Idea: SDF1100.001

ALMA Thesis Planner (automated pipeline)

Title: Identification, Spatial Distribution, and [CII]/FIR Energetics of the $z \approx 5.65$ SDF1100.001 Submillimeter Galaxy Group

Project Overview: This project utilizes pipeline-restored, science-ready archive image products from ALMA Band 7 dataset ADS/JAO.ALMA#2012.A.00028.S targeting the bright AzTEC-selected submillimeter galaxy field SDF1100.001. The dataset provides 1100×1100 arcsec continuum imaging and spectral line mapping over a $40'' \times 40''$ region ($\sim 240 \times 240$ kpc) across assuming a group redshift of $z = 5.65$ at $\sim 1''$ spatial resolution (~ 6 kpc) at $z = 5.65$ across the line redshift coverage $z = 5.63 - 5.72$ for rest-frame [CII] line emission.

Key Student Tasks: 1. **Source Identification and Spatial Mapping:** Ingest pipeline-restored [CII] line image cubes and 1100×1100 arcsec continuum maps from the ALMA Science Archive. Verify positions of the 7 candidate [CII] emitters previously discovered by SMA, search for fainter group members down to the ALMA sensitivity limit ($\sim 10 \times$ deeper than SMA), and map their 2D sky positions across the $40''$ field to characterize the group's spatial layout (e.g., testing for a filamentary distribution). 2. **1D Spectral Profiling and Kinematics:** Extract 1D spectra at source positions to determine [CII] line centroids, integrated line fluxes, and line widths (FWHM). Calculate line-of-sight velocity offsets (km s^{-1}) relative to $z = 5.65$ to assess the dynamic spread of line emitters within the group. 3. **Continuum Photometry and Energetics ($L_{\text{[CII]}}/L_{\text{FIR}}$):** Measure 1100×1100 arcsec continuum flux densities (or place 3σ upper limits) at target locations. Derive far-infrared luminosities (L_{FIR}) from single-band 1100×1100 arcsec continuum fluxes using a standard modified blackbody model under typical assumed dust parameters ($T_{\text{dust}} \approx 40-50$ K, dust emissivity index $\beta \approx 1.5-2.0$). Compute total [CII] line luminosities ($L_{\text{[CII]}}$) and evaluate the $L_{\text{[CII]}}/L_{\text{FIR}}$ ratios against the range $10^{-2.4}$ to $10^{-2.1}$ reported from prior SMA observations to investigate star-formation efficiency in an overdense environment less than 1 Gyr after the Big Bang.

Feasibility and Scope: By working directly with pipeline-restored archive image products and focusing on 1D spectral line extraction, 2D spatial positioning, and single-band continuum photometry, this project avoids complex raw visibility calibration while remaining scientifically meaningful and fully achievable by an undergraduate student within a single academic year using standard astronomical tools (e.g., CASA image utilities, Python/Astropy).